Forecasting Exchange Rates Using Gemini and Copilot

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Research question

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- Do AI tools (Copilot and Gemini) produce accurate exchange rate forecasts?
- Can they outperform simple benchmark models like Random Walk and AR(1)?

Objectives:

- To produce real-time, out-of-sample exchange rate forecasts.
- To evaluate how accurate the forecasts are.
- To compare AI forecasts with simple benchmark models

Motivation:

- Exchange rates are difficult to predict
- Benchmark models are hard to beat (Meese & Rogoff, 1983)

Methods, Model and Data

- **AI Tools:** Microsoft Copilot and Google Gemini (2 prompts each)
- **Benchmarks:** Random Walk (RW) and AR(1) model
- **Forecast Errors:** $e_{(t+h)} = y_{(t+h)} - \hat{y}_{(t+h)}$
- **Evaluation metrics:** AFE, RMSPE, MDE | Unbiasedness t-test | DMW test
- **Data:** FRED daily exchange rates | ~1,196 obs per currency for AR(1)
- **Currencies:** Euro, GBP, NOK, SEK, and CHF

Results

EUR, GBP, CHF Both tools perform well: AFE near zero, RMSPE small, unbiasedness not rejected

NOK & SEK — Copilot (Prompt 1) struggles: AFE = −1.36 for SEK; DMW p-values = 0.00 vs. both benchmarks

Prompt 2 (with baseline value) improves Copilot significantly

Gemini better directional accuracy under Prompt 1: MDE = 0.00 for SEK, NOK, CHF

Neither AI consistently beats RW or AR(1) benchmarks

Limitations of your results

- **Very short evaluation window (only 5 days)** — low statistical power for DMW and unbiasedness tests.

- **Prompt sensitivity:** results vary a lot depending on how the AI is prompted.

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Thank you !

References

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AI citation

- **AI Tools Used as Forecasters (Subject of Study):**
  - Microsoft Copilot was used to generate daily exchange rate forecasts via structured prompts
  - Google Gemini was used to generate daily exchange rate forecasts via the same prompts.